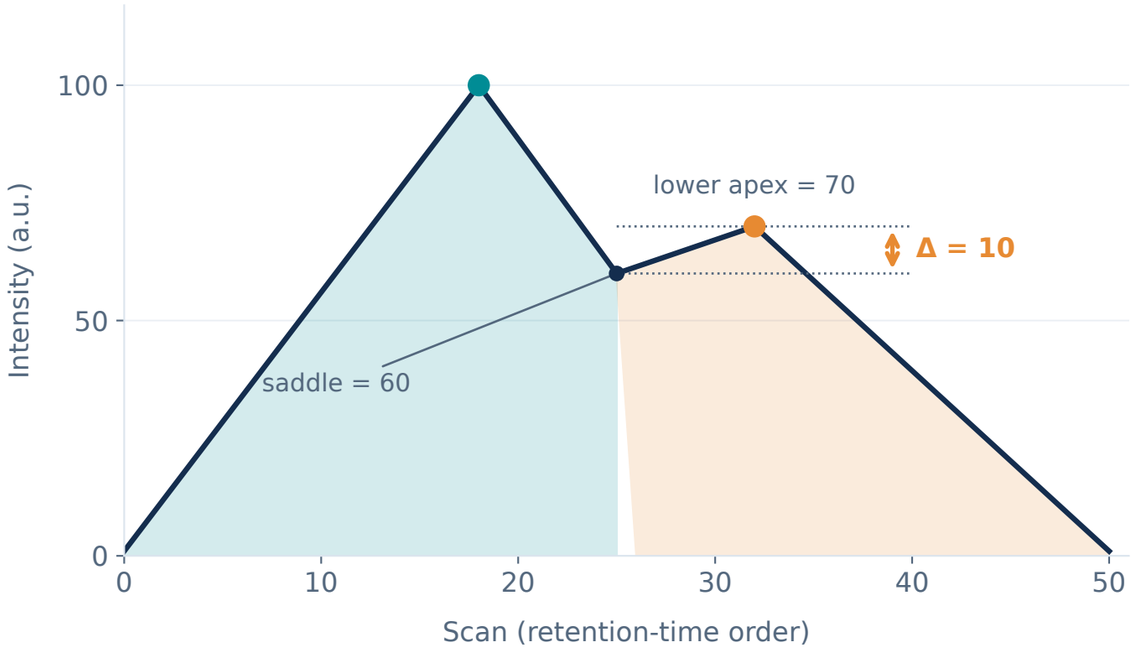
Should two neighbouring basins be one peak?

Persistence measures how far the lower apex rises above the connecting valley (saddle).

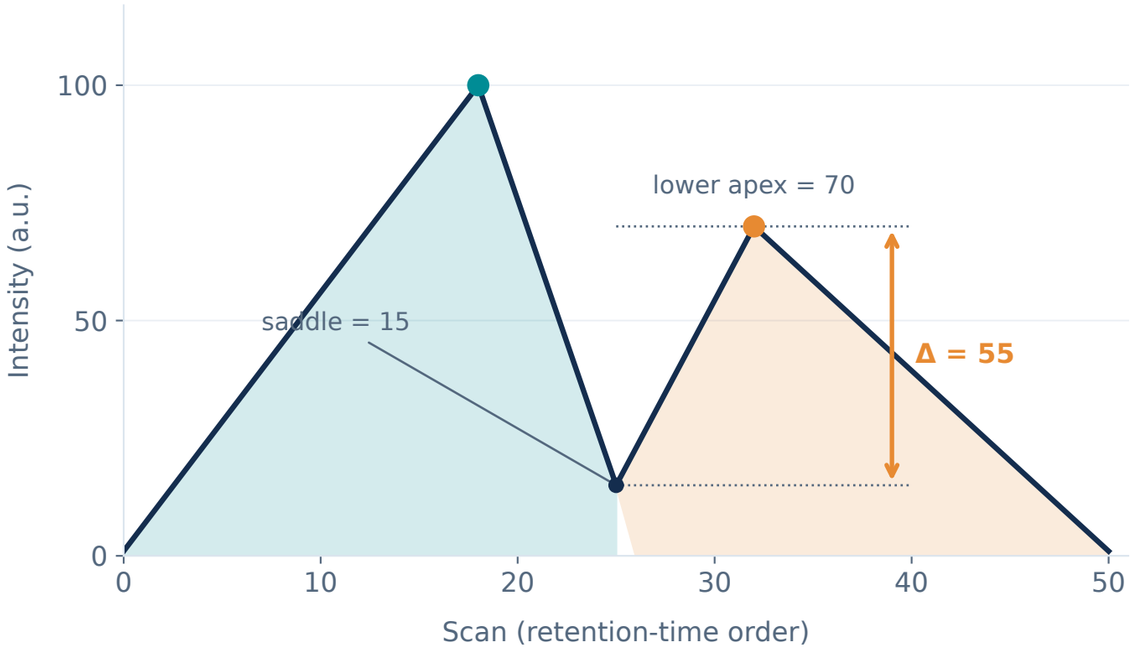
Shallow dip



MERGE → one candidate

Δ = 10 < threshold 35

Deep valley



RETAIN → two candidates

Δ = 55 ≥ threshold 35

Here: threshold = max(0.5 × lower apex, 3 × local noise) = 35

In these examples, a shallow dip merges; a deep, noise-resolved valley keeps two peaks.

Synthetic worked examples • default fractional merge rule • spacing and noise also affect the decision